

RH-067

Riemann Hypothesis Beta-Delta
Replay Obstruction Certificate

Beta8e / Delta-Relaxed High-Probe Audit

REPLAY OBSTRUCTION CHARACTERIZED
BETA-DELTA PARAMETER INTERFACE LOCK

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Integrity notice. This certificate records an audit result and proof-path dependency analysis. It does not claim that RH is proved or disproved, does not claim a shifted zero-free theorem, and does not claim that the original 7/8 theorem is false.

Executive Certificate

The audit certifies that the beta8e/delta-relaxed extension path does not replay with the original theorem inputs. It replays only after changing the input package to require shifted moments, an additional micro-smallness condition, and the final exponent weakened by exactly +8e, not +16e.

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1. Source theorem and interface point

The source theorem `actual_high_probe_from_raw_moments` assumes the old beta hypothesis and uses the downstream delta parameter shown below before calling `actual_nonfloor_rows_saving`.

Original beta hypothesis:

$$h\beta : (7/8 : \blacksquare) \leq \text{HeckeZeroSupremum.beta}$$

Original downstream delta parameter:

$$\Delta\blacksquare = \text{HeckeZeroSupremum.beta} - 7/8$$

Original moment parameter:

$$\text{SourceMomentsAt } \dots (\text{HeckeZeroSupremum.beta} - 7/8)$$

Under the original hypothesis, the old delta parameter is legitimate because $\beta \geq 7/8$ gives $0 \leq \beta - 7/8$. This certificate does not refute that original path.

2. Core obstruction statement

The beta8e replacement changes the usable beta condition to:

$$h\beta\blacksquare : 7/8 \leq \beta + 8e$$

That hypothesis does not imply:

$$0 \leq \beta - 7/8$$

Therefore the original downstream parameter cannot be reused under the beta8e hypothesis:

$$\Delta\blacksquare = \beta - 7/8 \text{ cannot be reused under } h\beta\blacksquare.$$

The required shifted parameter is:

$$\Delta\blacksquare = \beta + 8e - 7/8$$

3. Required repair inputs

A replay of the beta8e/delta-relaxed high-probe path requires changed theorem inputs:

Required shifted moment input:

$$\text{SourceMomentsAt } \dots (\beta + 8e - 7/8)$$

Required micro-smallness:

$$e \leq 1/800000$$

Final patched exponent:

$$\beta - 11/16 - \sigma + 8e$$

The shifted moment input is not cosmetic. The old source-visible input is at $\beta - 7/8$, and this audit established no source-visible monotonicity transfer from the old moment parameter to the shifted one.

4. Delta cap calculation

The relaxed delta cap is:

$$\Delta \leq 1/8 + 1/100000$$

For the shifted parameter:

$$\Delta_{\blacksquare} = \beta + 8e - 7/8$$

Using $\beta \leq 1$, the best direct bound is:

$$\begin{aligned}\Delta_{\blacksquare} &= \beta + 8e - 7/8 \\ &\leq 1 + 8e - 7/8 \\ &= 1/8 + 8e\end{aligned}$$

To obtain the micro-shift cap:

$$1/8 + 8e \leq 1/8 + 1/100000$$

one needs:

$$\begin{aligned}8e &\leq 1/100000 \\ e &\leq 1/800000\end{aligned}$$

5. Replay chain certified

The audit certifies the following replay chain at the level of proof-path dependencies:

Beta8e transport chain	actual_nonfloor_probe_transport_beta8e gives beta-side exponent $\beta - 11/16 - \sigma + 8e$.
Delta-relaxed row-saving chain	actual_nonfloor_rows_saving_deltaRelaxed preserves $3/16 + \Delta - \text{saving} + v + \text{dyadCost}$.
High-probe integration	Uses $\Delta_e = \beta + 8e - 7/8$.
Final comparison	Reduces to the original saving budget $\sigma + v + \text{dyadCost} \leq \text{saving}$.

6. Final exponent verification

The row-saving side contributes:

$$3/16 + \Delta_{\blacksquare} - \text{saving} + v + \text{dyadCost}$$

Substituting $\Delta_e = \beta + 8e - 7/8$:

$$\begin{aligned}3/16 + \Delta_{\blacksquare} &= 3/16 + \beta + 8e - 7/8 \\ &= \beta - 11/16 + 8e\end{aligned}$$

So the row-saving exponent becomes:

$$\beta - 11/16 + 8e - \text{saving} + v + \text{dyadCost}$$

The comparison to the patched transport exponent:

$$\beta - 11/16 - \sigma + 8e$$

reduces exactly to the original saving budget:

$$\sigma + v + \text{dyadCost} \leq \text{saving}$$

7. Final classification

Audit status	REPLAY OBSTRUCTION CHARACTERIZED
Obstruction type	BETA-DELTA PARAMETER INTERFACE LOCK
Original theorem directly refuted?	NO
Extension path replays with original inputs?	NO
Extension path replays after shifted inputs?	CONDITIONAL
Required new input	SourceMomentsAt($\beta + 8e - 7/8$)
Required new smallness	$e \leq 1/800000$
Final patched exponent	$\beta - 11/16 - \sigma + 8e$
Does +16e appear?	NO

8. What is not claimed

- This certificate does not claim that the original $7/8$ theorem is false.
- This certificate does not prove or disprove RH.
- This certificate does not prove a shifted zero-free theorem.
- This certificate does not show that the result is unrepairable in all possible formulations.
- This certificate does not claim an end-to-end Lean compiler verification.

It claims only:

The beta8e/delta-relaxed extension does not replay as stated with the original inputs.

9. Recommended public wording

Our audit identified a beta-delta interface lock in the proposed beta8e/delta-relaxed extension path. Under the replacement hypothesis $7/8 \leq \beta + 8e$, the old downstream parameter $\Delta = \beta - 7/8$ may be negative, so it cannot be reused. A replay requires the shifted parameter $\Delta_e = \beta + 8e - 7/8$, moments assumed at Δ_e , and a stronger smallness bound $e \leq 1/800000$. The final exponent becomes $\beta - 11/16 - \sigma + 8e$, with no +16e. This does not refute the original $7/8$ theorem, but it shows the relaxed extension does not replay with the original input package.

Document seal. Prepared by Grounded DI LLC / Mark S. Weinstein using MathWise DI². This certificate is intended as a formal audit memo, not as a claim of a completed end-to-end formal proof patch.